Explain the Linux file system hierarchy (/etc, /var/log, /home, /tmp) and file permissions: chmod, chown and SUID with examples.

Linux File System Hierarchy and File Permissions

Linux follows a hierarchical directory structure beginning with the root directory /.

Important Linux Directories

/ - Root directory

/etc - System configuration files

/var/log - System and application logs

/home - User home directories

/tmp - Temporary files

/usr - User-space programs and shared resources

/bin - Essential commands (often linked to /usr/bin)

Directory	Purpose	Example
/etc	Stores system configuration	/etc/passwd, /etc/ssh/sshd_config
/var/log	Stores system and application logs	/var/log/auth.log on many Debian-based systems
/home	Stores regular users' personal files	/home/student/
/tmp	Stores temporary files	/tmp/report.txt

Linux File Permissions

Linux uses three basic permission types:

Permission Symbol Numeric Value Meaning for a file

Read	r	4	View contents
Write	w	2	Modify contents
Execute	x	1	Execute as a program

Permissions are applied to three categories: Owner (u), Group (g), and Others (o).

For example-

-rwxr-xr—

- Owner: rwx = 7
- Group: r-x = 5
- Others: r-- = 4

Therefore, the numeric permission is 754.

A. chmod - Change File Permissions

The chmod command modifies file or directory permissions.

chmod 755 script.sh

This gives the owner read, write, and execute permissions, while the group and others receive read and execute permissions.

Another example:

chmod u+x script.sh

Adds execute permission for the file owner.

B. chown - Change Ownership

The chown command changes the owner or group of a file.

sudo chown student report.txt

Changes the owner of report.txt to student.

sudo chown student:developers report.txt

Changes the owner to student and the group to developers.

C. SUID - Set User ID

SUID is a special permission bit that allows an executable program to run with the effective privileges of the file owner rather than the user executing it.

Example- The passwd utility commonly uses SUID so ordinary users can change their own passwords while the utility performs protected operations.

Check its permissions:

ls -l /usr/bin/passwd

Example output-

-rwsr-xr-x 1 root root ... /usr/bin/passwd

The s in the owner's execute position indicates SUID.

To set SUID on an executable you own in an authorized lab:

chmod u+s example_program

Or:

chmod 4755 example_program

Security risk: If a privileged SUID executable contains a vulnerability, an attacker may be able to exploit it for privilege escalation. SUID should therefore be used only when necessary